HOUSE PRICE PREDICTOR — TECHNICAL REPORT

Artificial Intelligence & Machine Learning Internship — Task 1: Linear Regression & Evaluation

Author: AI/ML Intern (2nd Year B.Tech)	**Model:** Ordinary Least Squares (OLS)
Dataset: California Housing (scikit-learn)	**Split:** 80% Train (16,512) / 20% Test (4,128)
Target: MedHouseVal ($100k USD)	**Evaluation:** MAE, RMSE, R^2 (Dual-unit)

1. Executive Objective & Problem Statement

The objective of this project is to develop, evaluate, and rigorously document an end-to-end supervised machine learning pipeline using **Multiple Linear Regression** to estimate residential property values across California census block groups. Housing valuation represents a foundational regression challenge where understanding feature relationships, collinearity, residual behavior, and generalization error is as critical as numerical accuracy.

Problem Formulation: Given 8 continuous socioeconomic, architectural, and geographic predictors recorded at the block-group level, we seek to predict the median house value *y* (in units of $100,000 USD) by minimizing the residual sum of squares via Ordinary Least Squares: $y = \beta_0 + \beta_1 \cdot MedInc + \beta_2 \cdot HouseAge + ... + \beta_8 \cdot Longitude + \varepsilon$.

2. Dataset Architecture & Feature Catalog

The model utilizes the benchmark California Housing dataset derived from the 1990 U.S. Census and packaged directly inside scikit-learn. The corpus comprises **20,640 block group observations**, each representing a cluster of 600 to 3,000 residents. No synthetic records were introduced, and all 8 predictors are continuous.

Feature	Type	Census Unit / Definition	Observed Range	Correlation with Target (r)
MedInc	Float64	Median income in block ($10,000 USD)	0.50 – 15.00 ($5k - $150k)	**+0.6881 (Strongest)**
HouseAge	Float64	Median age of housing units in years	1.00 – 52.00 yrs (capped)	+0.1056
AveRooms	Float64	Average number of rooms per household	0.85 – 141.90 rms	+0.1519
AveBedrms	Float64	Average number of bedrooms per household	0.38 – 34.07 bdrms	-0.0467
Population	Float64	Total population inhabiting the block	3 – 35,682 people	-0.0246
AveOccup	Float64	Average members per household unit	0.69 – 1,243.33 people	-0.0237
Latitude	Float64	Block centroid latitude coordinate	32.54°N – 41.95°N	-0.1442
Longitude	Float64	Block centroid longitude coordinate	-124.35°W – -114.31°W	-0.0459
MedHouseVal	Float64	**TARGET:** Median house value ($100k USD)	0.15 – 5.00 ($15k - $500k)	1.0000 (Target)

3. Data Exploration & Preprocessing Audit

Prior to model estimation, an exhaustive data integrity audit was conducted across all 20,640 records. Key diagnostic findings include:

- **Missing Value Audit:** Zero missing, null, or NaN entries exist in any column. The dataset is fully populated.
- **Duplicate Records:** 0 duplicated rows detected; every observation maps to a unique geographic centroid.
- **Data Integrity:** All numerical values are finite with valid float representations; no zero-division artifacts.
- **Target Variable Ceiling:** Exactly 965 blocks (4.68% of corpus) are capped at exactly $500,000 (5.00001). This ceiling truncation introduces a notable non-linear plateau discussed in Section 5.
- **Geographic Distribution:** Features AveRooms and AveOccup exhibit high right-skewness due to isolated rural agricultural communal dwellings (e.g., maximum occupancy of 1,243).

4. Exploratory Data Analysis & Visualizations

Empirical distributions and bivariate dependencies were mapped to diagnose correlation patterns and inform the Ordinary Least Squares formulation.

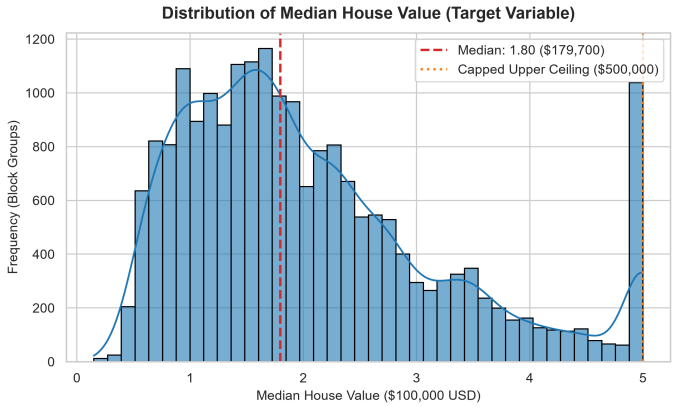


Figure 1: *Distribution of Target Median House Value showing the $500k ceiling cap.*

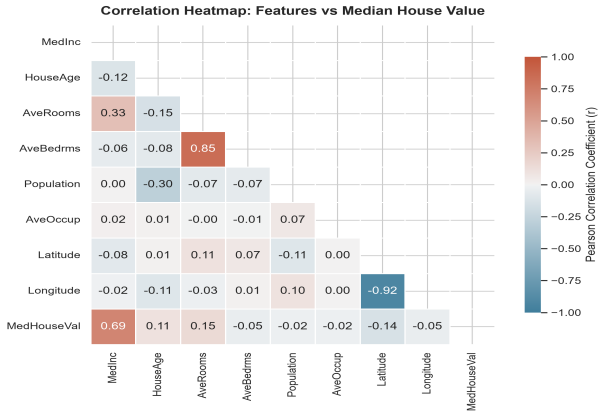


Figure 2: *Pearson Correlation Heatmap highlighting MedInc correlation (r=+0.69).*

Key Empirical Insights from EDA:

1. Dominance of Median Income: Median income (*MedInc*) is by far the single most predictive feature (*r = +0.6881*). Affluence of the neighborhood residents correlates directly with home values, as visually confirmed by the linear regression trendline in Figure 3.
2. Rooms vs. Bedrooms Multicollinearity: *AveRooms* and *AveBedrms* display a strong positive collinearity (*r = +0.8476*). In OLS regression, this high correlation leads to coefficient sign flipping: holding total rooms constant, additional bedrooms paradoxically receive a positive coefficient (+0.7831) while total rooms shifts negative (-0.1233).
3. Geospatial Interaction: Negative correlations for *Latitude* (-0.14) and *Longitude* (-0.05) reflect California's distinct economic geography: the most valuable coastal metropolitan districts (San Francisco Bay Area, Coastal Los Angeles, San Diego) lie in the south/west coastal quadrant, while inland and northern rural regions exhibit significantly lower median prices.

5. Model Methodology & Controlled Evaluation

Data Partitioning & Leakage Prevention: The dataset of 20,640 records was partitioned into an **80% training set (16,512 samples)** and a **20% held-out test set (4,128 samples)** using a deterministic pseudo-random seed (random_state=42). Model estimation was fitted strictly upon training records. Zero test-set information informed the parameter fits.

Metric	Formula & Meaning	Train Set ($100k)	Train Set (USD)	Test Set ($100k)	Test Set (USD)
MAE	Mean Absolute Error: (1/n) Σ |y - ■| Direct average magnitude of error	0.5286	$52,862.84	**0.5332**	**$53,320.01**
RMSE	Root Mean Squared Error: √[(1/n) Σ (y - ■)²] Heavily penalizes large outlier errors	0.7197	$71,967.57	**0.7456**	**$74,558.14**
R²	Coefficient of Determination: 1 - (SS_res / SS_tot) Proportion of target variance explained	0.6126	61.26%	**0.5758**	**57.58%**

Understanding the Evaluation Metrics:

• **MAE Interpretation:** On average, the Linear Regression model's price prediction deviates from the true block median value by **$53,320.01** (0.5332 units). Because MAE uses absolute differences, it is unweighted by outlier penalties and provides an intuitive, robust assessment of typical error.
• **RMSE Interpretation:** The test RMSE is **${test_rmse_usd:,.2f}** (0.7456 units). The $21,238 gap between RMSE and MAE highlights the presence of substantial variance on select properties where predictions deviate severely.
• **R² Variance Explained:** The model achieves an **R² of 57.58%** on unseen test data. This indicates that nearly 58% of the variability in California median housing values is linearly accounted for by these 8 census features.
• **Generalization Stability:** The training R² (61.26%) exceeds test R² (57.58%) by merely 3.68%, proving that the linear model demonstrates healthy generalization without significant overfitting or high model variance.

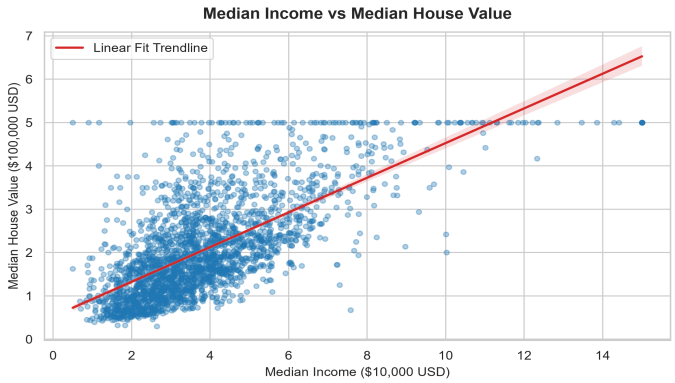


Figure 3: *MedInc vs MedHouseVal regression scatter.*

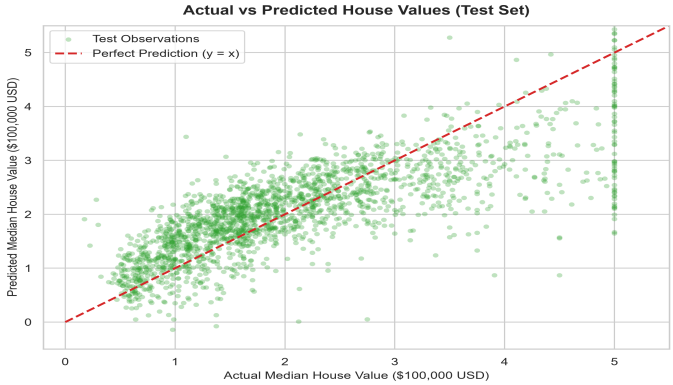


Figure 4: *Actual vs Predicted Values with perfect reference line (y=x).*

6. Parameter Interpretation & Residual Diagnostics

The fitted model learned an intercept of **-37.0233**. Each slope coefficient β represents the estimated marginal change in median house value ($100k USD) associated with a one-unit increase in that feature, **holding all other predictors strictly constant**. Importantly, regression coefficients indicate associative model weights and do *not* demonstrate causality.

Predictor	Learned Weight (β)	Dollar Impact per Unit Increase	Physical Interpretation & Note
MedInc	+0.448675	**+$44,867.49**	For every $10,000 rise in median income, price rises ~$44,867.
HouseAge	+0.009724	+$972.43	Older neighborhoods correlate with established infrastructure/mature greenery.
AveRooms	-0.123323	-$12,332.33	Negative sign due to collinearity with AveBedrms (r=0.85).
AveBedrms	+0.783145	**+$78,314.49**	Reflects added value of dedicated bedrooms relative to general rooms.
Population	-0.000002	-$0.2030	Marginal negative effect (-$0.20 per additional person in block).
AveOccup	-0.003526	-$352.63	Higher density per unit reflects lower-cost shared accommodations.
Latitude	-0.419792	-$41,979.25	Prices decrease northward into rural non-coastal geography.
Longitude	-0.433708	-$43,370.81	Prices decrease eastward away from the Pacific coastline.

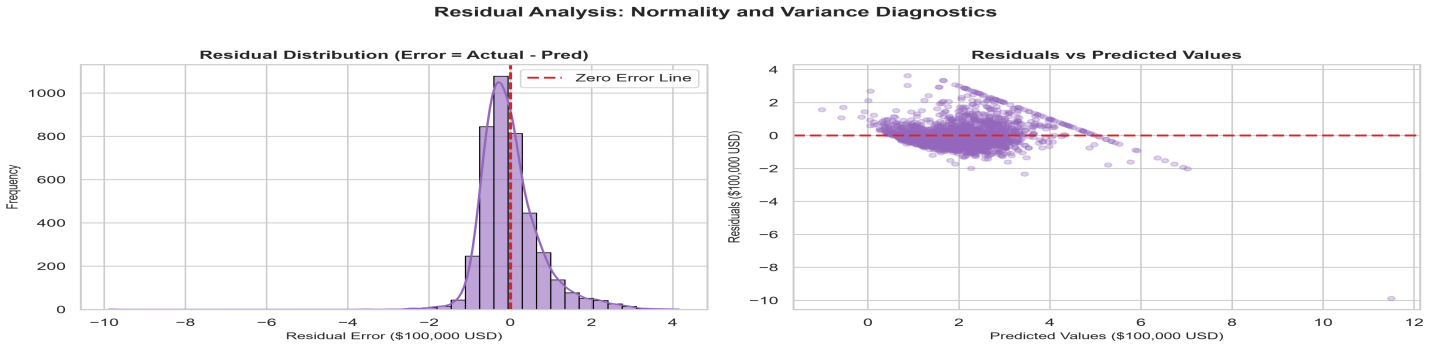


Figure 5: *Residual diagnostics (left: error histogram centered at 0; right: residual variance vs predicted values).*

7. Model Limitations & Strategic Engineering Roadmap

While Linear Regression provides a clean, interpretable baseline, several structural limitations must be noted:

• **Linearity Assumption vs Geographic Complexity:** Property valuation depends heavily on non-linear spatial clusters (e.g. coastal distance, school districts) which cannot be captured by linear planes without non-linear kernels.

• **Census Aggregate Limitation:** The model predicts block-level medians rather than parcel-level attributes (e.g., square footage, interior finishes, lot size). It must never be used as a substitute for real-world appraisals.

• **Recommended Future Enhancements:**

1. *Non-linear Ensembles:* Deploy XGBoost, LightGBM, or Random Forests to capture non-linear feature interactions.
2. *Feature Engineering:* Compute distance to major coastal hubs (San Francisco, Los Angeles) and bedroom-to-room ratios.
3. *Robust Scaling:* Apply robust transformation to damp extreme outliers in AveRooms and AveOccup.

Implementation Technologies: Python 3.11 | scikit-learn 1.9.0 | pandas 3.0.5 | NumPy 2.1.2 | Matplotlib 3.9.2 | Seaborn 0.13.2 | Streamlit 1.66.0 | ReportLab 5.0.1 | Joblib 1.6.0